



BAULKHAM HILLS HIGH SCHOOL

2012
YEAR 12 HALF-YEARLY

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions
- Start a new page for each question

Total marks – 70

Exam consists of 9 pages.

This paper consists of TWO sections.

Section 1 – Pages 4-5

Multiple Choice

Question 1-10 (10 marks)

Section 2 – Pages 5-8

Extended Response

Question 11- 14 (60 marks)

Standard integrals provided on page 9

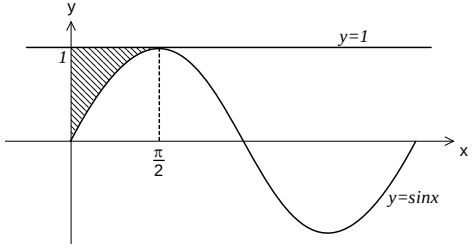
Section I - 10 marks

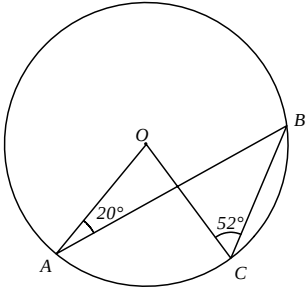
Attempt questions 1-10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for question 1-10

Marks

1)	A polynomial $P(x)$ has a relative maximum at $(-2,4)$, a relative minimum at $(1,1)$ and a relative maximum at $(5,7)$ and no other critical points. How many real <u>zeros</u> does $P(x)$ have? (A) one (B) two (C) three (D) four	1
2)	If $\frac{dy}{dx} = \cos x \cdot \sin^2 x$ then (A) $y = \sin^3 x + c$ (B) $y = \cos^2 x + c$ (C) $y = \sin^2 x + c$ (D) $y = \frac{1}{3}\sin^3 x + c$	1
3)	What are the solutions of the equation $\sin 2\theta = \cos \theta$ in the domain $-\pi \leq \theta \leq \pi$ (A) $\frac{\pi}{2}, \frac{3\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$ (B) $-\frac{\pi}{2}, \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$ (C) $-\frac{\pi}{6}, -\frac{5\pi}{6}, \frac{\pi}{6}, \frac{5\pi}{6}$ (D) $-\frac{\pi}{4}, \frac{\pi}{4}$	1
4)	Three boys and four girls are to sit around a table. How many arrangements are there if 2 specific girls sit next to each other? (A) $5!$ (B) $5! \times 2!$ (C) $6! \times 2!$ (D) $7! \times 2!$	1
5)	$\frac{d}{dx} \ln[\cos(2x)]$ is (A) $\frac{1}{\cos 2x}$ (B) $\frac{-\sin 2x}{\cos 2x}$ (C) $-2 \tan 2x$ (D) $2 \sec 2x$	1
6)	Let R be the region between the graphs of $y = 1$ and $y = \sin x$ from $x = 0$ to $x = \frac{\pi}{2}$ The volume of the solid obtained by revolving R about the x -axis is given by  (A) $\pi \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$ (B) $2\pi \int_0^{\frac{\pi}{2}} x \cdot \sin x \, dx$ (C) $\pi \int_0^{\frac{\pi}{2}} (1 - \sin x)^2 \, dx$ (D) $\pi \int_0^{\frac{\pi}{2}} 1 - \sin^2 x \, dx$	1
7)	The inverse function to $y = \frac{6}{2x-3} + 1$ is (A) $y = \frac{x+1}{2x-2}$ (B) $y = \frac{-x+1}{2x-1}$ (C) $y = \frac{3+3x}{2x-2}$ (D) $y = xy - \frac{3x+3}{2}$	1

8)	 In the diagram, O is the centre of the circle, $\angle OAB = 20^\circ$ and $\angle OCB = 52^\circ$. The measure of $\angle ABC$, in degrees is (A) 20 (B) 52 (C) 32 (D) 72	1
9)	The number of the solutions to the equation $x = 10 \cos x$ is (A) 3 (B) 5 (C) 6 (D) 7	1
10)	If $2x^3 - 9x^2 + 13x + k$ is divisible by $x - 2$, then it is also divisible by (A) $(x + 2)$ (B) $(x + 1)$ (C) $(x - 1)$ (D) $2x + 1$	1
End of Section 1		

Section II – Extended Response

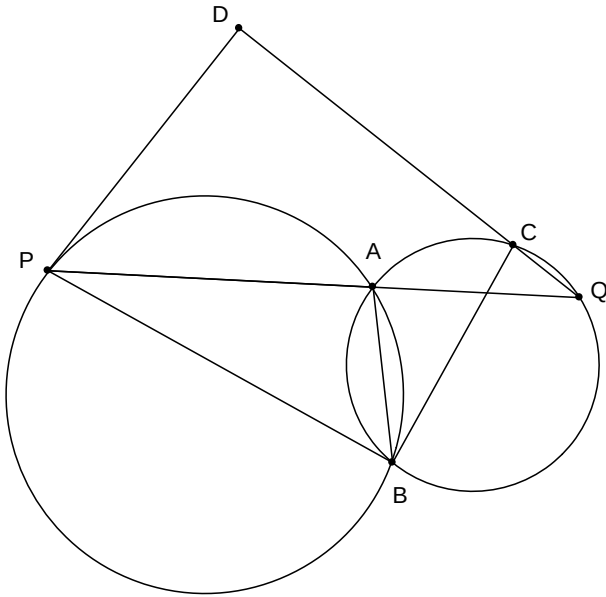
Attempt all questions. Show all necessary working.

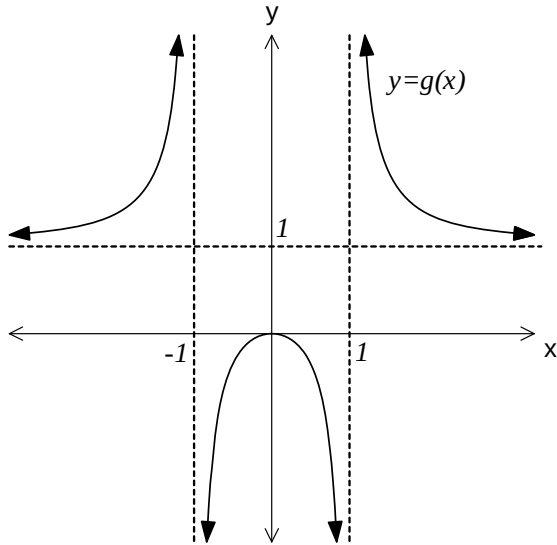
Start each question on a new page. Clearly indicate question number.

Write your name and teacher's name at the top of each new page.

Question 11 (15 marks) - Start a new page		Marks
a)	Solve for x: $\frac{2}{x-2} \geq 2x-1$	3
b)	Let A be the point $(-8, -3)$ and $B(4, 7)$ Find the coordinates of the point P that divides AB externally in the ratio 1: 2	2
c)	State the domain and range of the function $f(x) = 2 \sin^{-1} \left(\frac{x}{2} \right)$ and sketch it.	3
d)	Evaluate the integral below in exact form. $\int_{\frac{1}{2}}^{\frac{\sqrt{3}}{2}} \frac{1}{4x^2 + 1} dx$	2
e)	Prove that $\frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} = \sec \theta$	3
f)	How many five-person committees can be selected from 6 men and 8 women if there must be at least two women included?	2

Question 12 (15 marks) - Start a new page	Marks
a) i) Write $\sin x - \cos x$ in the form $A \sin(x - \alpha)$ where $0 \leq \alpha \leq \frac{\pi}{2}$ ii) Hence or otherwise solve the equation $\sin x - \cos x = 1$ for $0 \leq x \leq 2\pi$	2 2
b) When the polynomial $P(x)$ is divided by $x^2 - 5x + 6$, the remainder is $5x + 2$. What is the remainder when $P(x)$ is divided by $x - 2$?	2
c) A surveyor stands at the point P due south of a tower OT of height h, and finds the angle of elevation of the top of the tower to be 40°. Then he walks $100m$ to the point Q due east of the tower. The angle of elevation from Q to the top of the tower is then 30° i) Find the expressions for OP and OQ in terms of h ii) Show that $h = \frac{100(\tan 40^\circ \tan 30^\circ)}{\sqrt{\tan^2 40^\circ + \tan^2 30^\circ}}$ iii) Find h to the nearest metre.	1 3 1
d) The diagram shows the graph of the parabola $x^2 = 4ay$. PT is the tangent and PN is the normal at $P(2ap, ap^2)$. Given the coordinates for $T(ap, 0)$ and $N(0, a(p^2 + 2))$ i) Show that the equation of the tangent at P is $y = px - ap^2$ ii) M is the midpoint of TN, find the cartesian equation for the locus of M	2 2

Question 13 (15 marks) - Start a new page		Marks
a)	Use mathematical induction to prove that $16^n + 10n - 1$ is divisible by 25 for all integers $n \geq 1$	4
b)	 Two circles intersect at A and B. P is a point on the first circle and Q is a point on the second circle such that PAQ is a straight line. The tangent at P and the line QC produced (where C is on the second circle) meet at D i) Give a reason why $\angle DPA = \angle PBA$ 1 ii) Give a reason why $\angle CQA = \angle CBA$ 1 iii) Hence show that $BCDP$ is a cyclic quadrilateral 2	
c)	At time t years the number N of individuals in a population is given by $N = A + Be^{-t}$, where A and B are constants. After $\log_e 3$ years, there are 60 individuals and after $\log_e 5$ years there are 48 individuals. i) Find A and B 3 ii) Find the limiting population size. 1	
d)	Determine how many numbers can be formed using all of the digits 1,2,3, ... 9 without repetition in each of the following cases i) Even and odd digits alternate 1 ii) The digits 1,2,3 are together but not necessarily in that order 2	

Question 14 (15 marks) - Start a new page		
a)	Find $\int 6 \cos x (e^{3 \sin x}) dx$	2
b)	The function $g(x)$ is defined by $g(x) = \frac{x^2}{x^2-1}$ and its graph is shown below.  i) What is the largest domain of $g(x)$ including $x = 2$ for which an inverse function exist? 1 ii) Find the equation of the inverse function $g^{-1}(x)$ for this domain 2 iii) What is the domain of $g^{-1}(x)$? 1 iv) Find the value of $g^{-1}(g(-2))$ 1 v) Find the x-ordinates of all points of intersection of the two curves $y = g(x)$ and $y = g^{-1}(x)$ 2	
c)	Show that $\tan^{-1} 4 - \tan^{-1} \left(\frac{3}{5}\right) = \frac{\pi}{4}$ without the use of a calculator.	2
d)	The region bounded by $y = 2\cos^{-1} x$ and the x and y -axes is rotated around the y -axis. Find the volume of the solid generated.	4
End of Examination		

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

BAULKHAM HILLS HIGH SCHOOL
2012 HALF YEARLY
Mathematics Extension I
MULTIPLE CHOICE ANSWER SHEET

Section 1 - Answer Sheet

- | | | | | |
|-----|-------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1) | ☐ A | ☒ B | ☐ C | ☐ D |
| 2) | ☐ A | ☐ B | ☐ C | ☒ D |
| 3) | ☐ A | ☒ B | ☐ C | ☐ D |
| 4) | ☐ A | ☒ B | ☐ C | ☐ D |
| 5) | ☐ A | ☐ B | ☒ C | ☐ D |
| 6) | ☐ A | ☐ B | ☐ C | ☒ D |
| 7) | ☐ A | ☐ B | ☒ C | ☐ D |
| 8) | ☐ A | ☐ B | ☒ C | ☐ D |
| 9) | ☐ A | ☐ B | ☐ C | ☒ D |
| 10) | ☐ A | ☐ B | ☒ C | ☐ D |

Answers

Yr. 12 1/2 yearly 2012

Extension 1

Q. 11

$$a) \frac{2}{x-2} \geq 2x-1 \quad | \times (x-2)^2$$

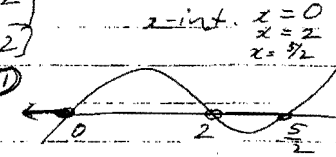
$$2(x-2) \geq (2x-1)(x-2)^2$$

$$0 \geq (x-2)[(2x-1)(x-2) - 2]$$

$$0 \geq (x-2)[2x^2 - 5x + 2 - 2]$$

$$0 \geq (x-2)[x(2x-5)] \quad (1)$$

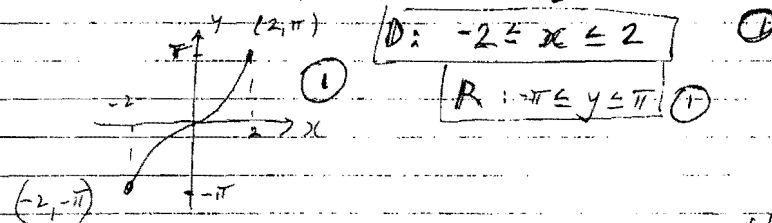
$$x \leq 0, \quad 2 \leq x \leq \frac{5}{2} \quad (1)$$



$$b) A(-8, -3) \quad B(4, 7)$$

$$P\left(\frac{-8x-2+4x+1}{1-2}, \frac{-3x-2+7x+1}{1-2}\right) = (-20, -13) \quad (1) \quad (1)$$

$$c) f(x) = 2 \sin^{-1}\left(\frac{x}{2}\right) \quad -1 \leq \frac{x}{2} \leq 1$$



$$d) \int_{1/2}^{1/2} \frac{1}{4x^2+1} dx = \int_{1/2}^{1/2} \frac{1}{4(x^2 + \frac{1}{4})} dx = \frac{1}{4} \int_{1/2}^{1/2} \frac{1}{x^2 + \frac{1}{4}} dx \quad (a^2 = \frac{1}{4})$$

$$= \frac{1}{4} \left[\frac{1}{\frac{1}{2}} \tan^{-1} 2x \right]_{1/4}^{3/4} = \frac{1}{2} \left[\tan^{-1} 2x \right]_{1/4}^{3/4} = \frac{1}{2} \left[\tan^{-1} \sqrt{3} - \tan^{-1} 1 \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{3} - \frac{\pi}{4} \right] = \frac{\pi}{24} \quad (1)$$

Q.11 cont.

$$\begin{aligned} \text{e) Simplify } \frac{\sin 2\theta}{\sin \theta} - \frac{\cos 2\theta}{\cos \theta} &= \frac{2\sin \theta \cos \theta}{\sin \theta} - \frac{\cos^2 \theta - \sin^2 \theta}{\cos \theta} \\ &= 2\cos \theta - \frac{2\cos^2 \theta - 1}{\cos \theta} = 2\cos \theta - 2\cos \theta + \frac{1}{\cos \theta} \\ &= \sec \theta = \text{RHS} \therefore \text{proven} \end{aligned}$$

f) 2W3M or 3W2M or 4W1M or 5W0M

$$\begin{aligned} &{}^8C_2 \cdot {}^6C_3 + {}^8C_3 \cdot {}^6C_2 + {}^8C_4 \cdot {}^6C_1 + {}^8C_5 \cdot {}^6C_0 \\ &\quad \quad \quad 560 \quad \quad \quad 840 \quad \quad \quad 420 \quad \quad \quad 56 \\ &= 1876 \quad \text{OR} \quad \frac{{}^{14}C_5}{2002} \cdot ({}^6C_5 + {}^6C_4 \cdot {}^8C_1) = 1876 \end{aligned}$$

Multiple choice

Question 1 B

2 D

3 B

4 B

5 C

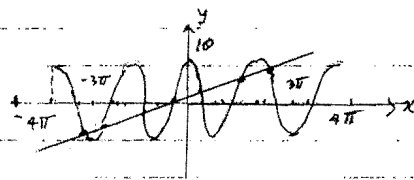
6 D

7 C

8 C

9 D

10 $k = -6 \therefore C$



Question 12

a) i) $\sin x - \cos x = A \sin(x - \alpha)$

$$A = \sqrt{1^2 + 1^2} = \sqrt{2} \quad \tan \alpha = \frac{1}{1} \therefore \alpha = 45^\circ = \frac{\pi}{4}$$

$$\therefore \sin x - \cos x = \sqrt{2} \sin(x - 45^\circ) = \sqrt{2} \sin(x - \frac{\pi}{4})$$

ii) $\sin x - \cos x = \sqrt{2} \sin(x - 45^\circ) = 1$
 $\sin(x - \frac{\pi}{4}) = \frac{1}{\sqrt{2}}$

$$\therefore x - \frac{\pi}{4} = \frac{\pi}{4}, \frac{3\pi}{4}$$

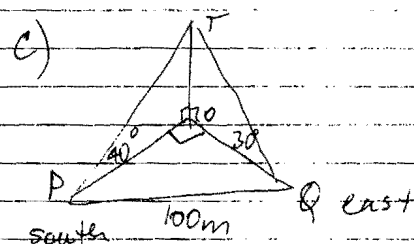
$$x = \frac{\pi}{2}, \pi$$

b) $P(x) \div (x^2 - 5x + 6) \therefore R(x) = 5x + 2$

$$\therefore P(x) = (x^2 - 5x + 6) \cdot Q(x) + 5x + 2$$

$$\therefore P(2) = (2^2 - 5 \cdot 2 + 6) \cdot Q(2) + 5 \cdot 2 + 2 = 12$$

$$\therefore R(2) = 12$$



i) $OP = \frac{OT}{\tan 40^\circ} = \frac{h}{\tan 40^\circ} \quad \tan 40^\circ = \frac{OT}{PO}$

$$OQ = \frac{OT}{\tan 30^\circ} = \frac{h}{\tan 30^\circ}$$

ii) ΔPOQ

$$100^2 = PO^2 + OQ^2$$

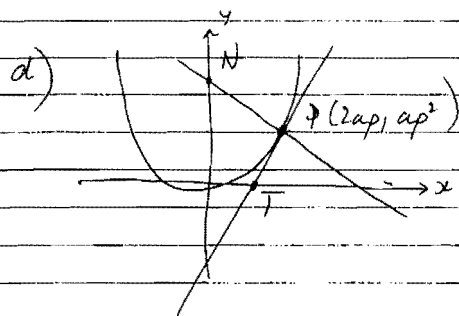
$$100^2 = \frac{h^2}{\tan^2 40^\circ} + \frac{h^2}{\tan^2 30^\circ}$$

$$100^2 = \frac{h^2 (\tan^2 30^\circ + \tan^2 40^\circ)}{\tan^2 40^\circ \cdot \tan^2 30^\circ}$$

$$\therefore h^2 = \frac{100^2 (\tan^2 40^\circ \cdot \tan^2 30^\circ)}{\tan^2 30^\circ + \tan^2 40^\circ} \therefore h = \frac{100 \cdot \tan 40^\circ \cdot \tan 30^\circ}{\sqrt{\tan^2 30^\circ + \tan^2 40^\circ}}$$

Q.12 c) cont.

iii) $h = 47.56... = 48 \text{ m (nearest metre)}$ ①



i) $x^2 = 4ay$: $y = \frac{x^2}{4a}$

$\frac{dy}{dx} = \frac{2x}{4a}$: at $x = 2ap$
 $m = \frac{2 \times 2ap}{4a} = p$ ①

tangent: $y - ap^2 = p(x - 2ap)$ ①
 $y = px - 2ap^2 + ap^2$
 $\therefore y = px - ap^2$

ii) $M_{TN} \left(\frac{ap}{2}, \frac{a(p^2+2)}{2} \right)$ ①
 $T(ap, 0)$
 $N(0, a(p^2+2))$

$\therefore$ locus of M $\begin{cases} x = \frac{ap}{2} \\ y = \frac{a(p^2+2)}{2} \end{cases}$ $p = \frac{2x}{a}$

$y = \frac{a}{2} \left(\left(\frac{2x}{a} \right)^2 + 2 \right) = \frac{a}{2} \left(\frac{4x^2}{a^2} + 2 \right)$

$y = \frac{2x^2}{a} + a$ ①

Question 13

a) $16^n + 10n - 1$ divisible by 25 $n \geq 1$

STEP 1: prove the expression is div. by 25 for $n=1$

$\therefore 16^1 + 10 \times 1 - 1 = 25$ which is div. by 25 $\therefore$ proven ①

STEP 2: assume that $16^k + 10k - 1$ is div. by 25

$\therefore 16^k + 10k - 1 = 25m$ ① (where m, k are integers)

STEP 3: prove that the expr. is div. by 25 for $n=k+1$

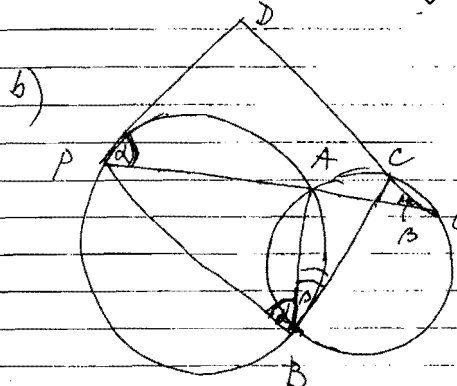
proof: $16^{k+1} + 10(k+1) - 1 = 16^k \cdot 16 + 10k + 10 - 1$

from assumption $-16^k = 25m - 10k + 1$
 $= (25m - 10k + 1) \cdot 16 + 10k + 9 = 25m \times 16 - 160k + 16 + 10k + 9$
 $= 25 \times 16m - 150k + 25 = 25(16m - 6k + 1)$

$\therefore$ divisible by 25. ① integer since $16m, 6k, 1$ are integers.

Conclusion: since the statement is proven to be true for $n=1$ and if it's true for $n=k$ and since proven for $n=k+1 \therefore$ true for $n=2, 3, \dots$ true for all $n \geq 1$ by induction.

if the conclusion missing ①



i) $\angle DPA = \angle PBA = \alpha$
 angle between tangent PD ① and a chord PA at the point of contact is $\equiv$ to the $\angle$ in alternate segment

ii) $\angle CBA = \angle CQA = \beta$
 (angles at the circumference in the same segment are equal)

Q. 13 b - cont.

iii) in $\triangle PDQ$ $\angle PDQ = 180^\circ - \angle DPA - \angle DQP$
 $= 180^\circ - \alpha - \beta$ (angle sum in triangle)
 but $\angle PBC = \angle PBA + \angle CBA = \alpha + \beta$ (adjacent angles)
 $\therefore \angle PDQ$ is supplementary to $\angle PBC$ (opposite $\angle$'s)
 $\therefore PBCD$ is cyclic quadrilateral (in cyclic quad are supplementary)

c) $N = A + Be^{-t}$

i) $60 = A + Be^{-\log_e 3}$ $\therefore 60 = A + Be^{-\log_e 3^{-1}}$ $\therefore 60 = A + B \cdot 3^{-1}$
 $48 = A + Be^{-\log_e 5}$ $\therefore 48 = A + Be^{-\log_e 5^{-1}}$ $\therefore 48 = A + B \cdot 5^{-1}$

$\therefore 12 = B(3^{-1} - 5^{-1})$
 $12 = B \times \frac{2}{15}$ $\therefore 90 = B$ $A = 30$

ii) $\lim_{t \rightarrow \infty} (A + Be^{-t}) = \lim_{t \rightarrow \infty} (30 + 90e^{-\infty}) = 30$

d) 1, 2, 3, ..., 9 4 even 5 odd

i) $5 \times 4 \times 3 \times 2 \times 1 = 5! = 120$ $\therefore 5! \cdot 4! = 2880$
 odd must be first

ii) $1, 2, 3, 4, 5, 6, 7, 8, 9$ $\therefore 7! \times 3! = 30240$

Question 14

a) $\int 6 \cos x (e^{3 \sin x}) dx = 2 \int 3 \cos x \cdot e^{3 \sin x} dx$
 $= 2e^{3 \sin x} + C$

b) i) $x \geq 0, x \neq 1$

ii) $g(x) = \frac{x^2}{x^2 - 1} = y$

$\therefore g'(x): x = \frac{y}{y^2 - 1}$ $xy^2 - x = y^2$
 $y(x - 1) = x$

$y = \pm \sqrt{\frac{x}{x - 1}}$

but since $D_g: x \geq 0, x \neq 1$ $R_{g^{-1}}$ is $y \geq 0, y \neq 1$

$\therefore y = \sqrt{\frac{x}{x - 1}}$

iii) $D_{g^{-1}}: (range of the original) x \geq 0, x > 1$

OR $\frac{x}{x - 1} \geq 0 \therefore x(x - 1) \geq 0$
 $\therefore x \geq 0, x > 1$

iv) $g^{-1}(g(2)) = g^{-1}\left(\frac{4}{3}\right) = \sqrt{\frac{12}{3}} = 2$

v) $g(x) = \frac{x^2}{x^2 - 1}$ intersect at $y = x$ with $g'(x)$

$\frac{x^2}{x^2 - 1} = x$
 $x^2 = x(x^2 - 1)$
 $0 = x[x^2 - 1 - x]$

$x = 0, x = \frac{1 \pm \sqrt{1 + 4}}{2}$
 $x = \frac{1 \pm \sqrt{5}}{2}$
 in domain

Q.14

(v) cont.

$$\textcircled{\text{OR}} \quad g(x) = \frac{x^2}{x^2-1} \quad \& \quad g'(x) = \sqrt{\frac{x}{x-1}}$$

$$\therefore \frac{x^2}{x^2-1} = \sqrt{\frac{x}{x-1}} \quad \textcircled{1}$$

$$\frac{x^4}{(x^2-1)^2} = \frac{x}{x-1}$$

$$x^4(x-1) = x(x-1)(x+1)^2$$

$$x(x-1)[x^3 - (x-1)(x+1)^2] = 0$$

$$x=0, \quad x=1, \quad x = \frac{1 \pm \sqrt{5}}{2} \quad \textcircled{1}$$

excluded from domain

$$c) \text{ Show } \tan^{-1} 4 - \tan^{-1}\left(\frac{3}{5}\right) = \frac{\pi}{4}$$

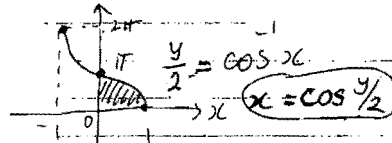
$$\text{LHS: } \tan\left(\underbrace{\tan^{-1} 4}_\alpha - \underbrace{\tan^{-1}\left(\frac{3}{5}\right)}_\beta\right) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \cdot \tan \beta} \quad \textcircled{1}$$

$$= \frac{4 - \frac{3}{5}}{1 + 4 \cdot \frac{3}{5}} = 1$$

$$\text{RHS} = \tan\left(\frac{\pi}{4}\right) = 1 = \text{LHS} \therefore \text{proven}$$

$$d) \quad y = 2 \cos^{-1} x \quad -1 \leq x \leq 1$$

$$\textcircled{\pi} \quad 0 \leq 2y \leq 2\pi$$



$$V = \pi \int_0^\pi \left(\cos \frac{y}{2}\right)^2 dy = \pi \int_0^\pi \left(\frac{1}{2} \cos y + \frac{1}{2}\right) dy \quad \textcircled{1}$$

$$= \frac{\pi}{2} \left[\sin y + y \right]_0^\pi = \frac{\pi}{2} [0 + \pi - 0 - 0] = \frac{\pi^2}{2} \quad \textcircled{1}$$

$$\cos y = 2 \cos^2 \frac{y}{2} - 1$$

$$\cos^2 \frac{y}{2} = \frac{1}{2} \cos y + \frac{1}{2}$$